The Four Fourier Representations

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Key Concepts

1) Fourier representations and analysis are named after Joseph Fourier, a French mathematician.

2) Fourier representations are based on expressing complicated signals as weighted "sums" of sinusoids of different frequencies.

3) Sinusoids are fundamental signals and very useful as building blocks.

    a) They describe physical phenomena in nature.

b) The notion of frequency is familiar and an intuitive description for the properties of signals.

c) LTI systems behave in a very special way with complex sinusoidal input signals.

4) There are four distinct Fourier representations for expressing a complicated signal as a weighted "sum" of sinusoids of different frequencies.

a) The Fourier transform (FT) applies to continuous-time, non periodic signals.

i) Frequency is continuous in the FT.

ii) The "sum" becomes an integral over frequency.

iii) The weights are a continuous function of frequency.

b) The Fourier series (FS) applies to continuous-time, periodic signals.

i) Frequency is discrete - integer multiples of the fundamental frequency of the signal.

ii) The weights are a discrete function of frequency.

c) The discrete-time Fourier transform (DTFT) applies to discrete-time non periodic signals.

i) Frequency is continuous in the DTFT.

ii) The "sum" becomes an integral over frequency.

iii) The weights are a continuous function of frequency.

iv) Discrete-time complex sinusoids are only unique over a 2π interval, so frequency is restricted to a 2π interval, often $(-\pi, \pi]$.

d) The discrete Fourier transform (DFT) describes discrete-time finite-duration (or periodic) signals.

i) Both frequency and time are discrete.

ii) Signal is described by N time samples or N DFT (frequency) coefficients.

iii) The fast Fourier transform (FFT) is a numerically efficient algorithm for computing the DFT. It is not a different representation.

5) The four Fourier representations have different roles in signal processing.

   a) The DFT is the only representation that can be numerically computed and thus is used extensively for computational applications.

   b) The DTFT is widely used for analysis of discrete-time systems and signal interactions.

c) The FS is used for analysis/synthesis of periodic signals and solving partial differential equations.

d) The FT is used for analysis of continuous-time signals and systems.

e) The FT and DTFT are used for analysis of mixtures of signal classes.

    i) The DTFT is used to analyze interactions between periodic and non periodic discrete-time signals.

    ii) The FT is used to analyze interactions between periodic and non periodic continuous-time signals.

iii) The FT is used to analyze interactions between continuous- and discrete-time signals.

Video Screenshots

Joseph Fourier 1768-1830

- Sums of sinusoids as solutions to heat transfer and vibration problems

Sinusoids are fundamental in nature

- electromagnetic spectum
- oscillatory motion

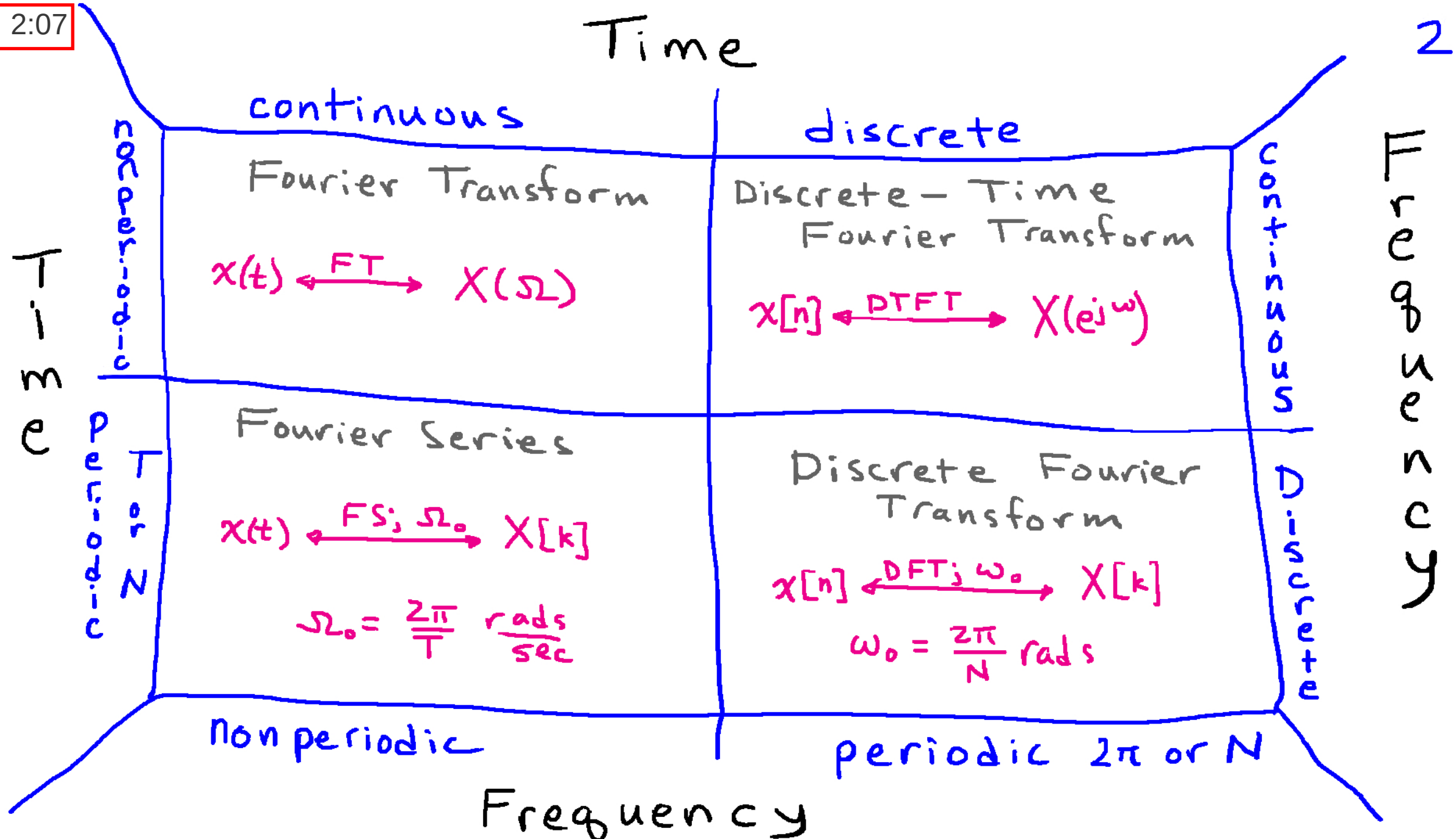
Notion of frequency is widespread

- sound

Sinusoids and LTI systems



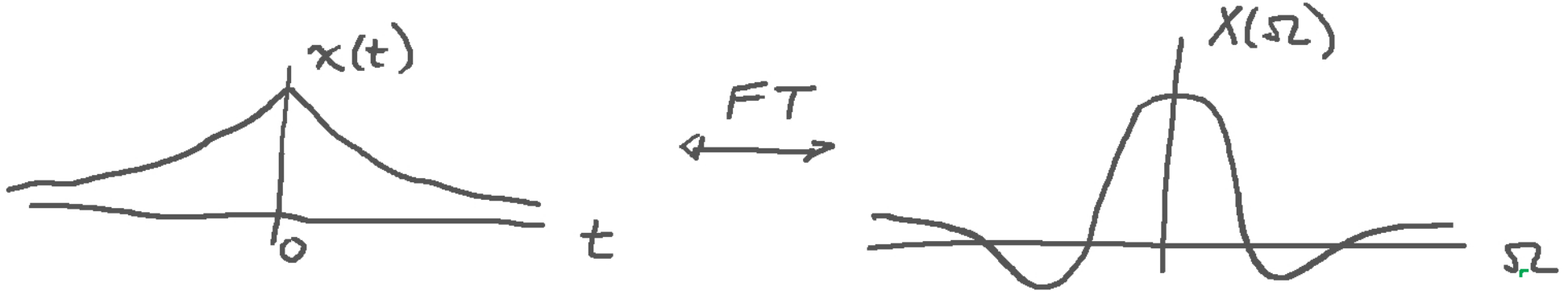
Sum of sinusoids can represent any signal!


Time
Frequency
Time
Frequency
continuous
discrete
non-periodic
periodic
Fourier Transform
$x(t) \xleftrightarrow{FT} X(\Omega)$
Discrete-Time Fourier Transform
$x[n] \xleftrightarrow{DTFT} X(e^{j\omega})$
Fourier Series
$x(t) \xleftrightarrow{FS;\ \Omega_0} X[k]$
$\Omega_0 = \frac{2\pi}{T}$ rads/sec
Discrete Fourier Transform
$x[n] \xleftrightarrow{DFT;\ \omega_0} X[k]$
$\omega_0 = \frac{2\pi}{N}$ rads
T or N
continuous
discrete
periodic
non periodic
periodic 2π or N

Fourier Transform

$$X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt \qquad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$$

$$\approx \sum_{\Omega} \underbrace{\frac{X(\Omega)}{2\pi} d\Omega}_{\text{weight}} \cdot e^{j\Omega t}$$



Discrete-Time Fourier Transform

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \qquad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega$$

$$x[n] \approx \sum_{-\pi \le \omega < \pi} \left(\frac{X(e^{j\omega})}{2\pi} d\omega\right) e^{j\omega n}$$

$$X(e^{j(\omega + \ell 2\pi)}) = X(e^{j\omega}) \quad \therefore \; X(e^{j\omega}) \text{ has period } 2\pi$$



Fourier Series: $x(t)$ is T-periodic

$$x(t+lT) = x(t)$$

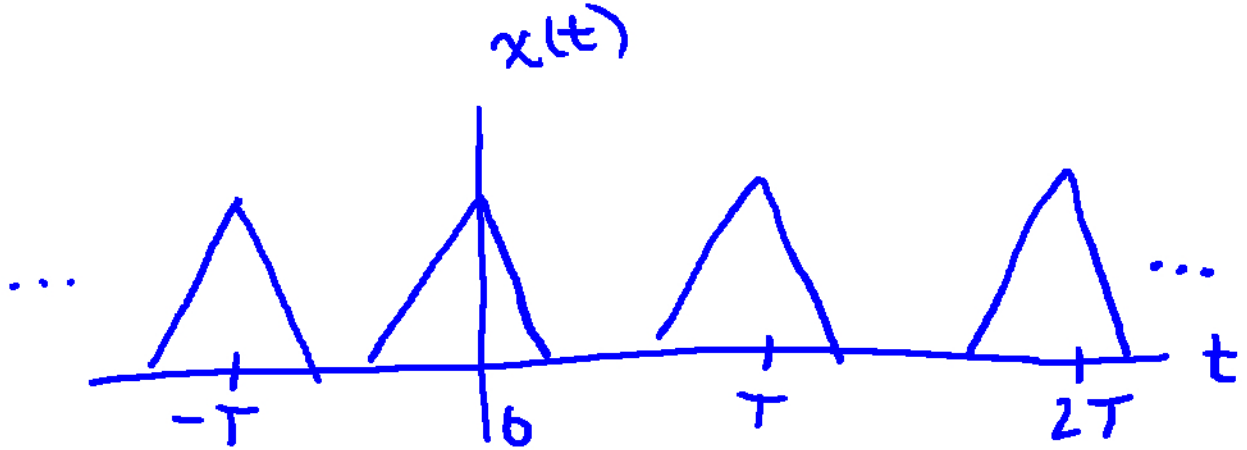
$$X[k] = \frac{1}{T}\int_0^T x(t)\, e^{-jk\Omega_0 t}$$

$$\Omega_0 = \frac{2\pi}{T}$$

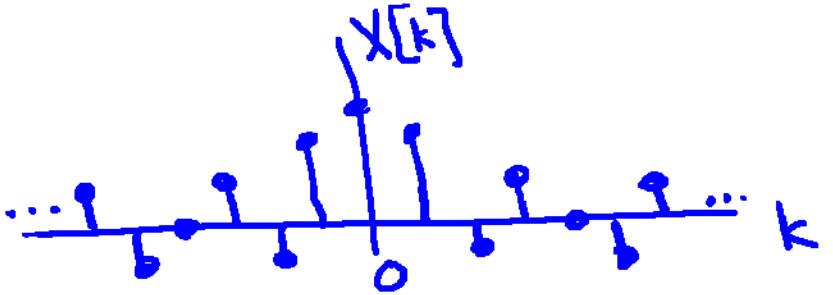
$$x(t) = \sum_{k=-\infty}^{\infty} X[k]\, e^{jk\Omega_0 t}$$

$X[k]$ weighting

$e^{jk\Omega_0 t}$ has frequency $k\Omega_0$



FS; Ω_0



Discrete Fourier Transform

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi kn}{N}} \qquad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi kn}{N}}$$

$x[n]$ N periodic

$X[k]$ N periodic

Only Fourier representation that can be computed exactly

FFT - fast algorithm for computing the DFT

Roles–

12:49

DFT: Computation!

DTFT: analysis of discrete-time systems and signals

FS: analysis/synthesis of periodic signals, solving PDEs

FT: analysis of continuous-time systems and signals, Analysis of mixtures of continuous- and discrete-time signals

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